

Study on an Intelligent Screening Method for Polycystic Ovary Syndrome Based on Deep PhysicsInformed Neural Network

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Abstract—Polycystic ovary syndrome (PCOS) not only causes anovulation in women but also severely affects their physical and mental health. Clinically, diagnostic delays often cause patients to miss optimal treatment windows. As a non-invasive detection technique, Raman spectroscopy has been used for screening this disease. In this letter, the Raman spectra of follicular fluid and plasma from women which PCOS are examined using a deep physics-informed neural network. The results demonstrate that by incorporating physical priors and integrating multi-domain spectral information, the proposed method achieves accuracies of 96.25% in detecting PCOS from plasma samples and 90.00% from follicular fluid samples.

Index Terms—Physicsinformed neural network, polycystic ovary syndrome, Raman spectroscopy, screening.

I. INTRODUCTION

POLYCYSTIC ovary syndrome (PCOS) is a common, complex and highly heterogeneous endocrinometabolic disorder of ovulation in women of reproductive age, with a prevalence of 5–20% in this group [1]. Besides reproductive dysfunction, patients often experience a range of complications—including pregnancyrelated complications, type II diabetes, cardiovascular disease and endometrial cancer—and significant psychological disorders, all of which seriously threaten women’s physical and mental health [2], [3].

At present, there are no highly specific biological markers for diagnosing PCOS; clinical judgment is mainly based on the 2003 Rotterdam consensus. However, this standard has obvious limitations: hormone levels are subject to fluctuation, ultrasound evaluation is subjective and the diagnostic thresholds are strict,

often leading to delayed diagnoses [4], [5]. Recently, emerging technologies such as genomics and metabolomics have opened new avenues for PCOS diagnosis, but their clinical application is hampered by high testing costs and limited diagnostic efficacy [6]. Thus, developing a noninvasive diagnostic method that combines high accuracy, low cost and simplicity of use is critical for early detection and precise intervention.

Raman spectroscopy provides a unique “fingerprint” of biomolecules via the inelastic scattering of light and has been widely adopted as a noninvasive diagnostic tool. Gulyamov and colleagues [7] first used Raman spectroscopy to analyse potential biomarkers associated with renal failure, suggesting that Raman spectroscopy has promise as a semiquantitative tool in chronic kidney disease. Liang and colleagues [8] combined Raman spectra of bonemarrow supernatant from different types of leukaemia with orthogonal partial least squares discriminant analysis to develop a noninvasive, simple method for subtype detection, achieving high specificity and sensitivity. Gao and colleagues [9] used Raman spectroscopy with multivariate statistics to propose a labelfree, noninvasive strategy for detecting and screening liver cancer cells. Cheng and colleagues [10] distinguished various hepatitis C patients by combining Raman spectra of blood with a support vector machine classifier. Zhang et al. [11] applied Raman spectroscopy and machinelearning techniques to distinguish breast cancer cells and subtypes, revealing molecular and structural differences between subtypes. Zhang et al. [12] also used Raman spectroscopy with machinelearning algorithms for PCOS diagnosis. Nogueira et al. [13] combined Fouriertransform infrared spectroscopy of saliva with a weighted knearest neighbour classifier to successfully classify healthy individuals, diabetes patients and periodontitis patients. Wu et al. [14] explored the use of surfaceenhanced Raman spectroscopy in diagnosing primary Sjögren’s syndrome and diabetic nephropathy. Cai et al. [15] applied a PCALDA algorithm with SERS to analyse urine samples from 87 coronaryheartdisease patients and healthy controls, achieving 90% diagnostic sensitivity. Robertson et al. [16] developed a COVID19 test based on urine Raman spectroscopy with an accuracy of 97.6%. Although Raman spectroscopy has achieved results in current disease diagnostics, existing methods mainly rely on traditional machinelearning algorithms; while these show good performance, they are sensitive to noise and prone to overfitting, limiting wider clinical use.

Raman spectroscopy faces signal complexity and significant background interference in disease diagnosis. In addition, large individual differences and limited samples further complicate

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analysis. To better characterise tissue properties, it is often necessary to collect large amounts of spectral data from each sample, increasing dataanalysis complexity and challenging traditional chemometric methods. Deep learning's datadriven nature makes it wellsuited to handle these issues: it can directly learn hidden features from large datasets and has succeeded in image recognition, including biomedical imaging [21], [22] [23]. Thanks to flexible architectures, deep learning has also been extended to onedimensional sequence data such as ECG and othr time-domain signals [24].

To fully exploit temporal dependencies and latent discriminative information in Raman spectra. A multimodal learning framework for non-invasive screening of PCOS proposed, which integrates signal and image domain features. Recurrence plots as a core technique, which maps raman signals into two-dimensional with nonlinear visual representations. And thereby provides interpretable structured inputs for deep neural networks by capturing the self-similarity structure in phase space.² Our main contributions are:

- 1) We introduce a deep physicsinformed learningbased screening method for PCOS that significantly improves discrimination capability in PCOS diagnosis using Raman spectra.
- 2) For PCOS Raman spectra, we propose a physicsinformation modelling method based on recurrence plots to better capture nonlinear dynamic features.
- 3) We design a dynamic pathselection, multisource feature-fusion module that enhances the adaptability and robustness of multimodal representation learning.

II. DATA AND PRE-PROCESSING

A. Data Sources and Spectral Acquisition

The data used in this study come from a publicly available Ramanspectroscopy dataset [12] of patients who underwent in vitro fertilization (IVF) in Shengjing Hospital, affiliated with China Medical University, including 50 PCOS patients (meeting the Rotterdam criteria) and 50 healthy controls. All subjects were free of endocrine diseases and recent hormone medication; blood and follicular fluid samples were collected and stored following standard procedures. Using a Basecare Raman 200 system (532 nm laser, 0.5 cm resolution), spectra in the 50–2000 cm^{-1} range were acquired. For each sample, five sites were measured with eight scans per site, producing 1024 data points per spectrum. The data processing method in this letter is the same as in [12].

B. Spectral Recurrence Plot

Recursion diagrams [21] [22], [23] as an time-dependent behavior of system dynamics have been widely used in the processing of medical time-series signals, such as heart rate variability, attention-deficit hyperactivity disorder classification, and cardiac arrhythmia distinction. These applications harnessed recursive plots transformed from time domain signals as deep learning inputs. Therefore, the Raman spectra are converted into 2D Raman encoding images as deep learning inputs for disease diagnosis. The conversion process is as follows

Suppose a wavenumber series $w = [w_1, w_2, w_3, \dots, w_T]$, the phase vector can be reconstructed by the Takens method as

$$\vec{w} = (w_i, w_{i+\tau}, \dots, w_{i+(m-1)\tau}), \forall i \in \{1, \dots, n - (m-1)\tau\}$$

where m is the embedding dimension and τ is the delay step. The recurrence matrix $R \in \mathbb{R}^{T \times T}$ is defined as

$$R_{i,j} = \Theta(\varepsilon - \|\vec{w}_i - \vec{w}_j\|), \forall i, j \in \{1, \dots, n - (m-1)\tau\}$$

where Θ is the Heaviside step function (equal to 1 when its argument is positive), ε is the distance threshold and $\|\cdot\|$ is a norm (commonly Euclidean). In this study, 1D Raman signals and 2D RP images as inputs for deep learning training and testing.

The resulting twodimensional matrix R can be visualised as a greyscale image—the recurrence plot—whose texture reveals periodicity, nonlinearity and dynamic characteristics of the time series.

III. METHOD

We propose a multimodal Ramanspectrum classification framework that fuses signaldomain and imagedomain features through pretraining and a dynamic path crossattention mechanism. The framework has two core stages: dualmodality pretraining and multisource feature fusion, as shown Fig. 1.

A. Dualmodality Pretraining

To fully utilise features in the temporal and image domains of Raman spectra, training begins with dualmodality pretraining, comprising signaldomain and imagedomain stages.

1) *Signaldomain Pretraining*: In the signal domain, the physical information neural network is used to learn the features of the original sequence of Raman spectra. To ensure that the model can take into account both the physical laws and statistical patterns of the spectral signal when extracting features, the model is constrained by a combination of Partial Differential Equation Residuals (PDE) and Cross-Entropy. The loss function at this stage can be expressed as

$$L = L_{ce} + \lambda L_{physics} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] + \lambda \frac{1}{M} \sum_{i=1}^M \|f(\hat{y}_i, y_i)\|^2$$

2) *Imagedomain Pretraining*: In the image domain, Raman spectra transformed into recurrenceplot images are processed by three convolutional networks: MobileViT [18], ResNet [19] and EfficientNet [20] (shown in the right of Fig. 1(b)). Each model learns image features independently during pretraining through Cross-Entropy constraints, preserving diversity and complementarity among architectures. Pretraining accelerates convergence in downstream multimodal tasks and enhances generalisation to different data distributions.

3) *Multisource Feature Fusion*: Traditional multisource fusion methods treat all pathways equally, which can introduce redundant or interfering information. We propose a dynamic pathselection fusion module that leverages heterogenous features extracted by multiple models and adaptively selects the most informative paths for fusion. The features extracted by the primary model (PINN) serve as query vectors, while features from auxiliary models act as keys and values. A crossattention module with dynamic path selection fuses them, enhancing features and supplementing semantics.

Given the input samples w , they are first extracted feature $F_w \in \mathbb{R}^{B \times L_q \times D}$ by the backbone model PINN, and then

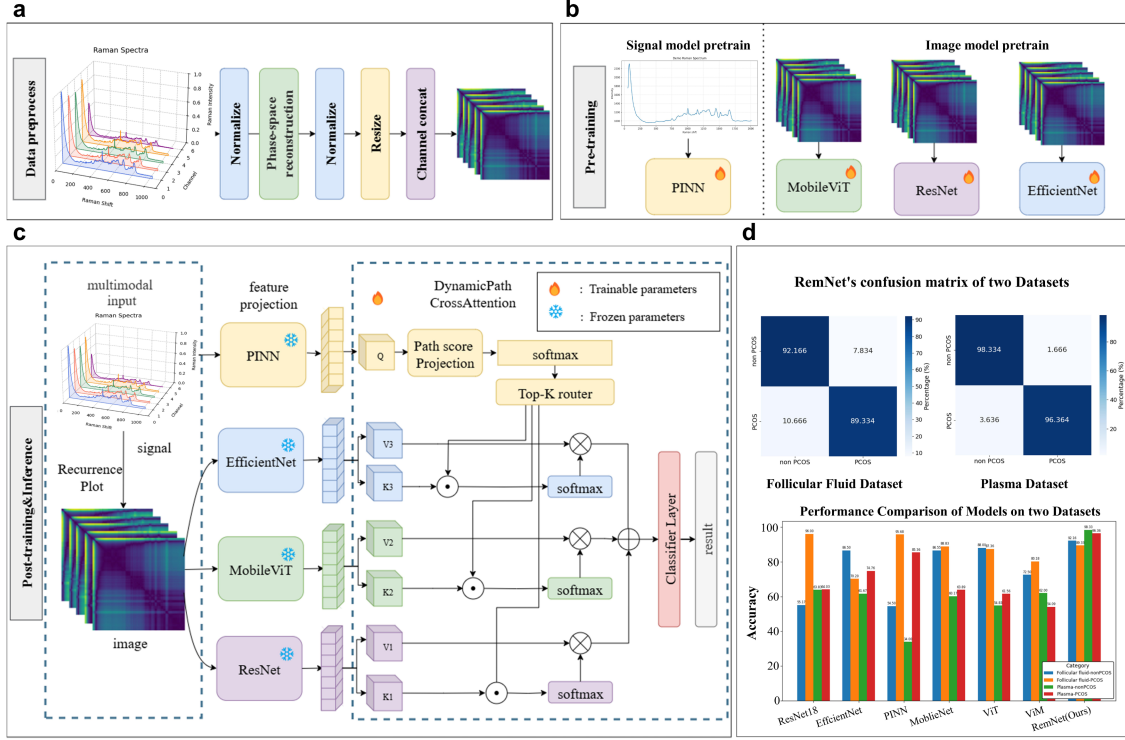


Fig. 1. Overall framework and experimental result analysis of the multimodal Raman spectroscopy classification model (a) Data preprocessing flow. The original Raman spectroscopy data is normalized, phase space reconstructed, image size unified, and channel concatenated, and finally transformed into multi-channel recursive graph images for downstream classification tasks. (b) Model pre-training stage. Firstly, PINN (Physical Information Neural Network) is used to pre-train the signal mode of Raman spectroscopy; then, MobileViT, ResNet, and EfficientNet of the image mode are pre-trained for image features. (c) Post-training and inference stage. The weights of PINN, MobileViT, ResNet, and EfficientNet are fixed, and only the dynamic path cross-attention module (DynamicPath CrossAttention) and the final classification layer are trained to achieve multimodal feature fusion and classification inference. (d) Experimental result display. The upper part shows the confusion matrix of the model on the test set, corresponding to the classification of follicular fluid and plasma; the lower part shows the comparison of classification accuracy of each model on the two types of samples (PCOS and non-PCOS).

mapped to a matrix through a linear transformation:

$$F_w = PINN(w) \quad Q = Linear_Q(F_w)$$

where, B represents the batch size, L_q indicates the length of the main output sequence, and D represents the embedding dimension.

Meanwhile, the three image domain models extract multiple sets of features respectively, where, and they are uniformly mapped to the same semantic space for fusion. They are respectively projected as keys and values. The mapping process is achieved through independent linear transformations

$$K_i = Linear_K^{(i)}(F_i), V_i = Linear_V^{(i)}(F_i)$$

This letter introduces a Path Scoring Network, implemented by a small multi-layer perceptron (MLP), which assigns weights to each path based on the global representation of the backbone features.

First, perform the pooling operation on to obtain the global representation:

$$q_{global} = \frac{1}{L_q} \sum_{j=1}^{L_q} Q_j$$

Then, the path scoring network is used to calculate the score for each path:

$$s = Softmax(MLP(q_{global})) \in \mathbb{R}^N$$

where, the “path” represents the importance of activation.

To further reduce the computational load, the Top-k sparse selection strategy is introduced. Only the paths with the top k highest scores are retained, and the corresponding weights of the remaining paths are set to 0:

$$\tilde{s}_i = \begin{cases} s_i, & i \in Top-k(s) \\ 0, & \text{otherwise} \end{cases}$$

The final path fusion weights are obtained through normalization $\tilde{s}_i$.

For the selected path, perform cross-attention calculations between the corresponding and the main trunk

$$A_i = softmax\left(\frac{QK_i^T}{\sqrt{D}}\right)$$

$$O_i = A_i V_i$$

Then integrate all the selected attention outputs

$$O = \sum_{i=1}^N \tilde{s}_i \cdot O_i$$

Finally, the fused features $O \in \mathbb{R}^{B \times L_q \times D}$ can be input into the classifier to obtain the final classification result.

TABLE I
COMPARISON OF CLASSIFICATION PERFORMANCE OF EACH MODEL ON FOLLICULAR FLUID AND PLASMA TEST SETS

Datasets	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Plasma	ResNet18	66.25	60.91	64.03	61.18
	MobileNet	61.25	60.95	48.44	53.38
	EfficientNet	65.00	64.89	74.76	65.81
	PINN	63.42	56.55	87.35	68.00
	ViT	55.00	56.82	61.56	54.65
	ViM	57.50	60.78	44.26	48.11
	REMNet(ours)	96.25	96.00	96.36	95.78
Follicular fluid	ResNet18	71.25	64.81	96.00	74.54
	MobileNet	86.25	80.36	88.83	83.30
	EfficientNet	78.75	76.10	70.20	72.43
	PINN	75.00	68.57	89.02	75.65
	ViT	80.00	82.78	74.86	75.56
	ViM	75.00	69.87	80.18	72.47
	REMNet(ours)	90.00	87.11	89.33	87.12

IV. EXPERITMENTS AND RESULTS

To validate the performance of the proposed model (REMNet) in multichannel Ramanspectra classification, we conducted comparative and ablation experiments evaluating classification accuracy, perclass performance, pathselection effects and model interpretability.

A. Experimental Settings and Evaluation Metrics

1) *Dataset Configuration*: Five Ramanspectral channels were collected per sample, each containing a 5×1024 intensity sequence. After normalisation, each channel was converted to a 224×224 recurrenceplot image and concatenated to form a $5 \times 224 \times 224$ multichannel image. The data were split 4/1 into training and testing sets, with two classes (PCOS and NonPCOS).

2) *Model Settings and Training Parameters*: The REMNet model uses a lightweight PINN as the “query path” encoder and employs ResNet18, EfficientNet and MobileViT as three feature paths in the dynamic fusion module. A staged pretraining strategy was used. The Adam optimiser with an initial learning rate of $1e4$, batch size of 32 and total of 200 epochs was adopted, with early stopping based on validation loss. All experiments applied five-fold cross-validation and ran on an NVIDIA A100 GPU. The effectiveness of the proposed method was evaluated using accuracy, precision, recall and F1 score.

B. Comparative Experiments

To validate the effectiveness of REMNet, we compared with current methods: PINN, ResNet18, EfficientNet, MobileViT, Vision Transformer (ViT) [25] and Vision Mamba(ViM) [26]. Results in Table I showed that on the plasma test set, REMNet achieved 96.25% accuracy and an F1 score of 95.78%, outperforming all baseline models. On the follicularfluid test set, REMNet achieved 90.00% accuracy with perfect precision and recall, whereas other models reached 70–86% accuracy. REMNet shows especially strong recall, indicating better identification of positive samples and fewer false negatives.

C. Ablation Experiments

To further assess the dynamic pathscoring module, we performed ablation studies on a mixed dataset of follicularfluid and

TABLE II
RESULTS OF ABLATION EXPERIMENT

Model	Accuracy(%)	F1-Score(%)
REMNet w/o path score	90.30	89.80
REMNet w/o soft Attention	91.60	91.50
REMNet(ours)	92.80	92.70

TABLE III
COMPARISON OF CALCULATION EFFICIENCY BETWEEN REMNET AND VARIOUS BASELINE MODELS

Model	Parameters (M)	VARAM (MB)	Times (ms)	Accuracy (%)
PINN	0.99	17.42	12.27	82.5
ResNet18	11.18	61.05	22.34	87.5
EfficientNet	4.01	39.17	32.47	85.0
MobileNet	0.95	19.26	35.29	85.0
ViT	86.19	466.6	54.58	80.0
ViM	7.05	304.43	177.16	75.0
REMNet(ours)	51.56	150.03	59.03	94.5

plasma samples. A version of REMNet without path scoring averaged features across all paths; another version normalised weights without Topk sparse selection; the full REMNet used complete path scoring and Topk activation with crossattention. Results in Table II showed that the full version achieved the highest accuracy, confirming that dynamic path scoring and sparse selection significantly improve fusion efficiency and suppress redundancy.

D. Inference Efficiency and Memory Consumption

Comparisons of computational efficiency in Table III show that although REMNet’s parameter count is higher than lightweight models like ResNet18, EfficientNet and MobileViT, its multipath fusion and dynamic scoring considerably enhance feature extraction and discrimination on multichannel Raman spectra. REMNet’s memory usage and average inference time are higher, but this cost yields superior classification performance, making it suitable where high diagnostic accuracy is essential.

V. CONCLUSION

This work presents a deep physicsinformed learningbased screening method for polycystic ovary syndrome using multisource Raman spectra. Physicsinformed learning deeply extracts potential features in Raman spectra, while dynamic path crossattention and pretraining strategies fuse multisource data. Using plasma and follicularfluid data, the proposed method achieves diagnostic accuracies of 96.25% and 90.00%, respectively. Ablation experiments show the proposed module improves fusion efficiency and suppresses redundant noise. Although the method modestly increases computational cost, it significantly enhances model expressiveness and discrimination, making it suitable for medical scenarios that require high diagnostic accuracy.

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